

Rabina Lama

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PROFESSIONAL SUMMARY

Senior Data Engineer with 8+ years of experience architecting, modernizing, and operating enterprise-scale cloud-native data platforms, with deep specialization in **Microsoft Azure**. Proven expertise in designing scalable lakehouse and analytics ecosystems using **Azure Data Lake Storage (ADLS)**, **Azure Databricks (Spark)**, **Azure Synapse Analytics**, **Azure Data Factory**, and **Event Hubs**, enabling reliable, high-performance batch and real-time data processing across insurance, healthcare, and financial services domains. Experienced in leading large-scale cloud transformation initiatives, re-architecting legacy data systems into governed, cost-optimized, and compliance-aligned Azure platforms that support enterprise analytics, executive reporting, actuarial modeling, regulatory submissions, and operational intelligence. Strong background in dimensional modeling, CDC-driven ingestion frameworks, distributed Spark optimization, and enterprise-grade ELT design patterns aligned with modern data product principles. Skilled in implementing secure and audit-ready cloud environments using **Azure Active Directory RBAC**, encryption with **Key Vault**, infrastructure-as-code with **Terraform**, and CI/CD automation through **Azure DevOps** pipelines. Adept at building observability-driven production systems leveraging **Azure Monitor** and logging frameworks to ensure reliability, performance optimization, and SLA adherence. Recognized for providing technical leadership, mentoring engineering teams, and partnering cross-functionally with analytics, compliance, actuarial, and business stakeholders to deliver trusted, scalable, and business-aligned Azure data platforms.

TECH STACK

- **Cloud Platform (Azure):** Azure Synapse Analytics, Azure Data Lake Storage (ADLS Gen2), Azure Databricks, Azure Data Factory, Event Hubs, Azure Functions, Azure SQL Database, Key Vault, Azure Monitor, Log Analytics
- **Data Engineering:** Spark, PySpark, Databricks Workflows, Kafka, Event Hub Streaming, CDC, ETL/ELT, Batch & Real-Time Pipelines
- **Data Warehousing:** Azure Synapse, Snowflake, Star/Snowflake Schema, Dimensional Modeling
- **Databases:** PostgreSQL, MySQL, SQL Server, Azure SQL Database, Cosmos DB
- **DevOps / IaC:** Terraform, Docker, Kubernetes (AKS), Azure DevOps, Jenkins, GitHub Actions
- **Monitoring / Observability:** Azure Monitor, Log Analytics, Datadog, Prometheus
- **Languages:** Python, SQL, Java, Scala, Bash

CERTIFICATIONS

- Microsoft Certified: Azure Solutions Architect Expert (AZ-305)
- Databricks Lakehouse Fundamentals

PROFESSIONAL EXPERIENCE

Senior Data Engineer

JP Morgan Chase, New York, NY

Feb 2024 - Present

- Led enterprise-wide modernization by migrating underwriting and claims analytics into a fully governed **Microsoft Azure ecosystem**, redesigning ingestion, transformation, and reporting layers to improve scalability, resilience, and long-term cloud cost efficiency.
- Architected a centralized enterprise data lake on **Azure Data Lake Storage (ADLS Gen2)** integrated with **Azure Synapse Analytics**, enabling standardized onboarding of multi-terabyte datasets across policy, billing, actuarial, and servicing domains.
- Built distributed batch processing frameworks using **Azure Databricks Spark**, implementing partition optimization, autoscaling, and workload tuning strategies that reduced end-to-end processing time by over 40%.
- Developed near real-time ingestion pipelines leveraging **Event Hubs streaming**, Spark Structured Streaming, and CDC-based workflows, reducing reporting latency from 48 hours to under 3 hours.
- Optimized executive and enterprise reporting workloads in **Azure Synapse** through clustering, partition strategies, materialized views, and SQL refactoring, accelerating dashboard performance while lowering overall compute cost.

- Designed enterprise ELT transformation frameworks using **Azure Data Factory**, **Python**, and Databricks-based dimensional modeling to ensure consistent business metrics and automated data quality validation.
- Provisioned production-grade cloud infrastructure using **Terraform** and fine-grained **Azure AD RBAC**, enabling compliant deployments across dev, QA, and production environments aligned with financial regulatory standards.
- Established CI/CD automation pipelines using **Azure DevOps** and GitHub workflows, reducing manual deployment effort and improving engineering governance across enterprise data services.
- Implemented observability and incident monitoring using **Azure Monitor**, Log Analytics, and alert-driven remediation workflows, improving pipeline uptime and reducing incident response time by more than 35%.
- Partnered closely with business, compliance, and analytics leadership teams to publish governed, audit-ready datasets supporting risk analytics, regulatory reporting, and enterprise decision-making.
- **Technologies Used:** Microsoft Azure (Synapse Analytics, ADLS Gen2, Databricks, Data Factory, Event Hubs, Key Vault, Azure Monitor), Spark, PySpark, Python, SQL, Terraform, Azure DevOps, GitHub Actions

Senior Data Engineer

Johnson & Johnson, Denver, CO

Mar 2022 – Jan 2024

- Architected an enterprise-scale **Azure lakehouse platform** leveraging **Azure Synapse Analytics**, **ADLS Gen2**, and **Azure Databricks**, enabling analytics-ready clinical and pharmaceutical datasets across multiple regulated business units.
- Developed high-throughput ingestion and transformation pipelines using **Spark on Azure Databricks**, processing structured and semi-structured clinical trial and manufacturing data to support compliance reporting and research analytics.
- Implemented real-time streaming architectures with **Event Hubs** and Spark Structured Streaming, improving supply chain visibility and proactive issue detection across manufacturing systems.
- Optimized enterprise analytics workloads in **Azure Synapse** through partition tuning, query acceleration, workload isolation, and SQL refactoring, reducing query latency by 40% for executive dashboards.
- Orchestrated batch and streaming workflows using **Azure Data Factory** and Databricks Workflows, reducing manual operational overhead while improving pipeline reliability and recoverability.
- Designed dimensional data marts and governed semantic layers in **Synapse**, enabling standardized KPIs for clinical research, regulatory compliance reporting, and executive-level healthcare dashboards.
- Applied HIPAA-aligned governance controls including **Azure AD policies**, encryption with **Key Vault**, audit logging, and secure dataset segmentation to protect sensitive healthcare information.
- Provisioned scalable cloud infrastructure using **Terraform** and automated deployments via **Azure DevOps**, ensuring reproducible environment consistency across dev and production.
- Implemented centralized observability using **Azure Monitor** dashboards and alert-driven runbooks to proactively detect failures, bottlenecks, and pipeline disruptions.
- Partnered with analytics, data science, and compliance teams to publish governed datasets as reusable Azure data products, strengthening audit readiness and accelerating enterprise lakehouse adoption.
- **Technologies Used:** Microsoft Azure (Synapse Analytics, ADLS Gen2, Databricks, Data Factory, Event Hubs, Key Vault, Azure Monitor), Spark, PySpark, Python, SQL, Terraform, Azure DevOps

Data Engineer

UnitedHealth Group (Optum), Minnetonka, MN

Jan 2021 – Feb 2022

- Designed scalable healthcare analytics pipelines leveraging **GCP Cloud Storage** and distributed Spark frameworks, supporting claims processing and population health analytics across millions of member records while maintaining strong data governance.
- Built ingestion workflows combining **Pub/Sub** streaming patterns and scheduled batch processing to improve operational reporting timeliness and reduce latency in eligibility, provider, and member analytics use cases.
- Developed transformation pipelines using **Python** and Spark to standardize high-volume healthcare datasets, enabling risk adjustment analytics, cost optimization initiatives, and improved downstream reporting consistency.
- Optimized analytical workloads in **BigQuery** through query tuning, partitioning, clustering strategies, and efficient join patterns, improving dashboard responsiveness by 30% for operational and clinical analytics consumers.
- Implemented automated validation and reconciliation checks to ensure claims dataset accuracy and completeness, reducing discrepancies in reporting outputs and increasing trust with actuarial and clinical stakeholder teams.

- Orchestrated workflows using **Cloud Composer (Airflow)**, improving automation coverage, reducing manual operational effort by 35%, and strengthening end-to-end pipeline reliability across daily processing cycles.
- Applied HIPAA-aligned controls using **IAM RBAC**, encryption, and audit logging to protect PHI across cloud-based datasets while supporting secure access patterns for analytics and reporting teams.
- Enhanced monitoring and alerting with **Cloud Monitoring** to reduce downtime, improve pipeline visibility, and accelerate incident response for critical reporting and eligibility pipelines.
- Collaborated with actuarial and clinical analytics teams to deliver trusted, analysis-ready datasets that powered predictive risk scoring initiatives and supported operational planning decisions.
- Supported infrastructure provisioning and repeatable deployments using **Terraform**, enabling secure configuration management and standardized environment setup across engineering teams.
- Improved reliability of ingestion services through robust error handling, retry patterns, and automated recovery workflows, reducing recurring load failures and improving SLA adherence.
- Documented pipeline logic, governance practices, and operational runbooks to improve maintainability and enable effective knowledge transfer across distributed engineering teams.
- **Technologies Used:** Google Cloud Platform (BigQuery, Cloud Storage, Pub/Sub, Cloud Composer, Cloud Monitoring), Spark, PySpark, Python, SQL, Terraform, Airflow

Backend Developer

Allstate Insurance Company, Chicago, IL

Jun 2019 – Dec 2021

- Developed Python-based backend services to automate ingestion and standardization of policy and claims datasets into enterprise analytics platforms, improving accessibility and usability for actuarial, reporting, and operational stakeholders.
- Built RESTful APIs and integration services using **Python** and **SQL**, exposing curated insurance datasets for compliance reporting, internal analytics products, and business intelligence applications.
- Developed scalable validation and transformation modules to cleanse, standardize, and enrich high-volume transaction records prior to downstream warehouse loading, improving data consistency and reducing reporting discrepancies.
- Engineered backend workflows integrating multiple upstream systems, reducing manual reconciliation effort and improving reliability of daily and weekly dataset refresh cycles across policy servicing operations.
- Optimized SQL transformation logic by improving indexing strategies, execution plans, and query refactoring, reducing batch processing latency by 30% while supporting growing data volumes.
- Designed relational database schemas in PostgreSQL to support scalable storage and retrieval of claims events used for historical trend analysis, operational reporting, and regulatory submissions.
- Implemented scheduled batch jobs using Python automation scripts and job orchestration patterns, ensuring timely delivery of refreshed datasets to dashboards and downstream consumer applications.
- Developed audit logging and monitoring features to improve traceability, data lineage visibility, and compliance readiness across ingestion and reporting services used by business stakeholders.
- Collaborated with engineers, analysts, and reporting teams to translate business requirements into reliable backend services, curated datasets, and reusable transformation components.
- Assisted in modernizing legacy flat-file workflows into scalable API-driven and event-driven integrations, improving pipeline stability and reducing processing failures across ingestion services.
- Improved pipeline resilience through robust error handling, retry mechanisms, and monitoring enhancements, reducing recurring failures by 20% and strengthening operational reliability.
- Maintained technical documentation of ingestion processes, schema changes, and service dependencies to improve cross-team knowledge sharing and long-term maintainability.
- **Technologies Used:** Python, SQL, PostgreSQL, REST APIs, Batch Processing, Linux, Git

EDUCATION

Bachelor's in Computer Science

University of Houston-Downtown, Houston, TX